
LECTURE NOTES

Discrete Mathematics notes

Functions

DM549 - Discrete Mathematics

Simon Holm
Fall 2024



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

1 Functions	3
1.1 Injective, Surjective, Bijective	3
1.2 Combining functions	3
1.3 Increasing, Decreasing, and more	4
1.4 Cardinality	4

1 Functions

Definition

Let A and B be non-empty sets. A function f from A to B , for each $x \in A$, assigns **precisely** one element $f(x) \in B$ to x .

$$f : A \rightarrow B$$

Where A is the domain and B is the co domain

Definition: Image (range)

Let $f : A \rightarrow B$ be a function. The *image* or *range* of f is

$$\text{im}(f) = \{f(x) \mid x \in A\} = \{y \in B \mid \exists x \in A : f(x) = y\}$$

1.1 Injective, Surjective, Bijective

Definition: Injective function

A function $f : A \rightarrow B$ is called injective or *one-to-one* if

$$\forall x_1, x_2 \in A : (f(x_1) = f(x_2) \implies x_1 = x_2),$$

that is f assigns ant value $y \in B$ to at most one $x \in A$

Definition: Surjective function

A function $f : A \rightarrow B$ is called surjective or *onto* if

$$y \in B \mid \exists x \in A : f(x) = y$$

that is, $\text{Im}(f) = B$

Definition: Bijective function

A function $f : A \rightarrow B$ is called bijective or *one-to-one correspondence* if ot is both injective and surjective.

1.2 Combining functions

Definition: Addition and multiplication of two functions

Let $f : A \rightarrow B$ and $g : A \rightarrow B$ be functions. Then $(f + g) : A \rightarrow B$ and $(f \cdot g) : A \rightarrow B$ are functions with

$$(f + g)(x) = f(x) + \frac{g(x)}{f \cdot g}(x) = f(x) \cdot g(x)$$

$\forall x \in A$

Definition: Composition of two functions

Let $f : A \rightarrow B$ and $g : A \rightarrow B$ be functions. Then the *composition* of g and f , $(f \circ g) : A \rightarrow B$ is a functions with

$$(g \circ f) = g(f(x))$$

$\forall x \in A$

Note that for composition the codomain of f has to match the domain of g

1.3 Increasing, Decreasing, and more

Definition

Let $f : A \rightarrow B$ If, for all $x_1, x_2 \in A$ with $x_1 < x_2$, it holds that

- $f(x_1) \leq f(x_2)$, f is called *increasing*
- $f(x_1) < f(x_2)$, f is called *strictly increasing*
- $f(x_1) \geq f(x_2)$, f is called *decreasing*
- $f(x_1) > f(x_2)$, f is called *strictly decreasing*

If f is increasing or decreasing, it is called monotone.

Observe that if a function f is either strictly increasing or strictly decreasing, it must also be injective.

1.4 Cardinality

Definition: Cardinality

Two sets A, B have the same cardinality ($|A| = |B|$), if there exists a bijection function from $A \rightarrow B$